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Java SE 27 & JDK 27

Class ThreadLocal<T>

java.lang.Object
    java.lang.ThreadLocal<T>

Type Parameters:

T - the type of the thread local's value

Direct Known Subclasses:

InheritableThreadLocal

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public class **ThreadLocal<T>**
extends Object

This class provides thread-local variables. These variables differ from their normal counterparts in that each thread that accesses one (via its `get` or `set` method) has its own, independently initialized copy of the variable. `ThreadLocal` instances are typically private static fields in classes that wish to associate state with a thread (e.g., a user ID or Transaction ID).

For example, the class below generates unique identifiers local to each thread. A thread's id is assigned the first time it invokes `ThreadId.get()` and remains unchanged on subsequent calls.

```
import java.util.concurrent.atomic.AtomicInteger;

public class ThreadId {
    // Atomic integer containing the next thread ID to be assigned
    private static final AtomicInteger nextId = new AtomicInteger(0);

    // Thread local variable containing each thread's ID
    private static final ThreadLocal<Integer> threadId =
        new ThreadLocal<Integer>() {
            @Override protected Integer initialValue() {
                return nextId.getAndIncrement();
        }
    };

    // Returns the current thread's unique ID, assigning it if necessary
    public static int get() {
        return threadId.get();
    }
}
```

Each thread holds an implicit reference to its copy of a thread-local variable as long as the thread is alive and the `ThreadLocal` instance is accessible; after a thread goes away, all of its copies of thread-local instances are subject to garbage collection (unless other references to these copies exist).

Since:

1.2

Constructor Summary

Constructors

Constructor	Description
`ThreadLocal()`	Creates a thread local variable.

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
`T`	`get()`	Returns the value in the current thread's copy of this thread-local variable.
`protected T`	`initialValue()`	Returns the current thread's "initial value" for this thread-local variable.
`void`	`remove()`	Removes the current thread's value for this thread-local variable.
`void`	`set(T value)`	Sets the current thread's copy of this thread-local variable to the specified value.
`static <S> ThreadLocal<S>`	`withInitial(Supplier<? extends S> supplier)`	Creates a thread local variable.

Methods declared in class Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Constructor Details

ThreadLocal

```
public ThreadLocal()
```

Creates a thread local variable.

See Also:

`withInitial(java.util.function.Supplier)`

Method Details

initialValue

```
protected T initialValue()
```

Returns the current thread's "initial value" for this thread-local variable. This method will be invoked the first time a thread accesses the variable with the `get()` method, unless the thread previously invoked the `set(T)` method, in which case the `initialValue` method will not be invoked for the thread. Normally, this method is invoked at most once per thread, but it may be invoked again in case of subsequent invocations of `remove()` followed by `get()`.

Implementation Requirements:

This implementation simply returns `null`; if the programmer desires thread-local variables to have an initial value other than `null`, then either `ThreadLocal` can be subclassed and this method overridden or the method `withInitial(Supplier)` can be used to construct a `ThreadLocal`.

Returns:

the initial value for this thread-local

See Also:

`withInitial(java.util.function.Supplier)`

withInitial

```
public static <S> ThreadLocal<S> withInitial
(Supplier<? extends S> supplier)
```

Creates a thread local variable. The initial value of the variable is determined by invoking the `get` method on the `Supplier`.

Type Parameters:

`S` - the type of the thread local's value

Parameters:

`supplier` - the supplier to be used to determine the initial value

Returns:

a new thread local variable

Throws:

`NullPointerException` - if the specified supplier is null

Since:

1.8

get

```
public T get()
```

Returns the value in the current thread's copy of this thread-local variable. If the variable has no value for the current thread, it is first initialized to the value returned by an invocation of the `initialValue()` method.

Returns:

the current thread's value of this thread-local

set

```
public void set(T value)
```

Sets the current thread's copy of this thread-local variable to the specified value. Most subclasses will have no need to override this method, relying solely on the `initialValue()` method to set the values of thread-locals.

Parameters:

`value` - the value to be stored in the current thread's copy of this thread-local.

remove

```
public void remove()
```

Removes the current thread's value for this thread-local variable. If this thread-local variable is subsequently read by the current thread, its value will be reinitialized by invoking its `initialValue()` method, unless its value is set by the current thread in the interim. This may result in multiple invocations of the `initialValue` method in the current thread.

Since:

1.5

Report a bug or suggest an enhancement
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